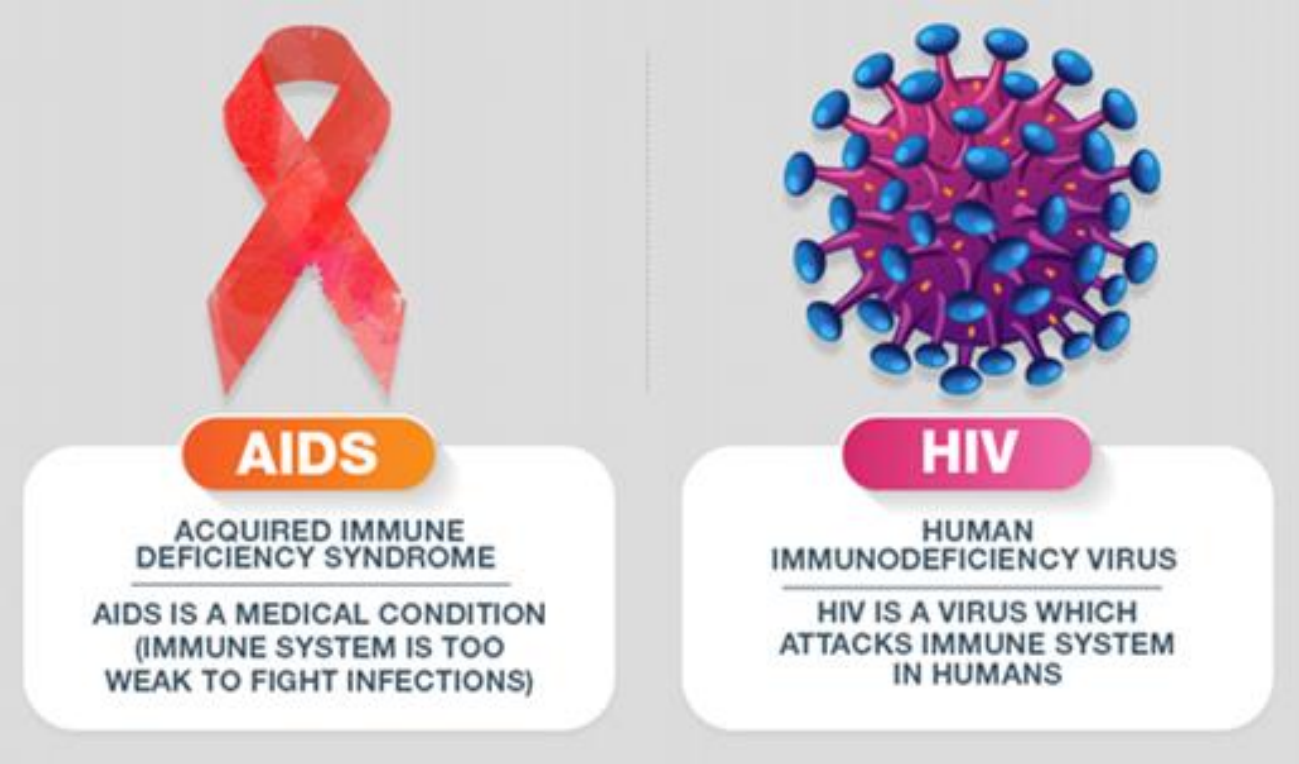
ACQUIRED IMMUNE DEFICIENCY SYNDROME (AIDS)



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INTRODUCTION

1. AIDS was first described in 1981 as a new disease.
2. In 1983, the causative agent was identified as Human Immunodeficiency Virus (HIV).
3. HIV is a retrovirus. It is classified into:
   a. HIV-1 (most common and virulent)
   b. HIV-2 (less virulent, mainly in West Africa)
4. HIV primarily attacks CD4+ helper T cells.
5. Leads to: Progressive immune system destruction
6. Increased susceptibility to: Opportunistic infections, Malignancies
7. According to Joint United Nations Programme on HIV/AIDS: Approximately 34 million people are living with HIV/AIDS.
8. High prevalence regions: Sub-Saharan Africa, Asia, South America
9. **High-Risk Groups**
   - Individuals with: Multiple sexual partners , STIs
   - Intravenous drug users
   - Healthcare workers (occupational exposure)
   - Infants born to HIV-positive mothers

Modes of Transmission

HIV is transmitted through **three main routes**:

Sexual Transmission

1. Most common mode.
2. Includes: Receptive anal intercourse , Vaginal intercourse
3. Risk increases with: Existing sexually transmitted infections (STIs):
    1. Syphilis
    2. Gonorrhea
    3. Chlamydia
    4. Herpes simplex
    5. Trichomoniasis
    6. Chancroid
4. Condom use reduces risk by ~20-fold

Parenteral Transmission

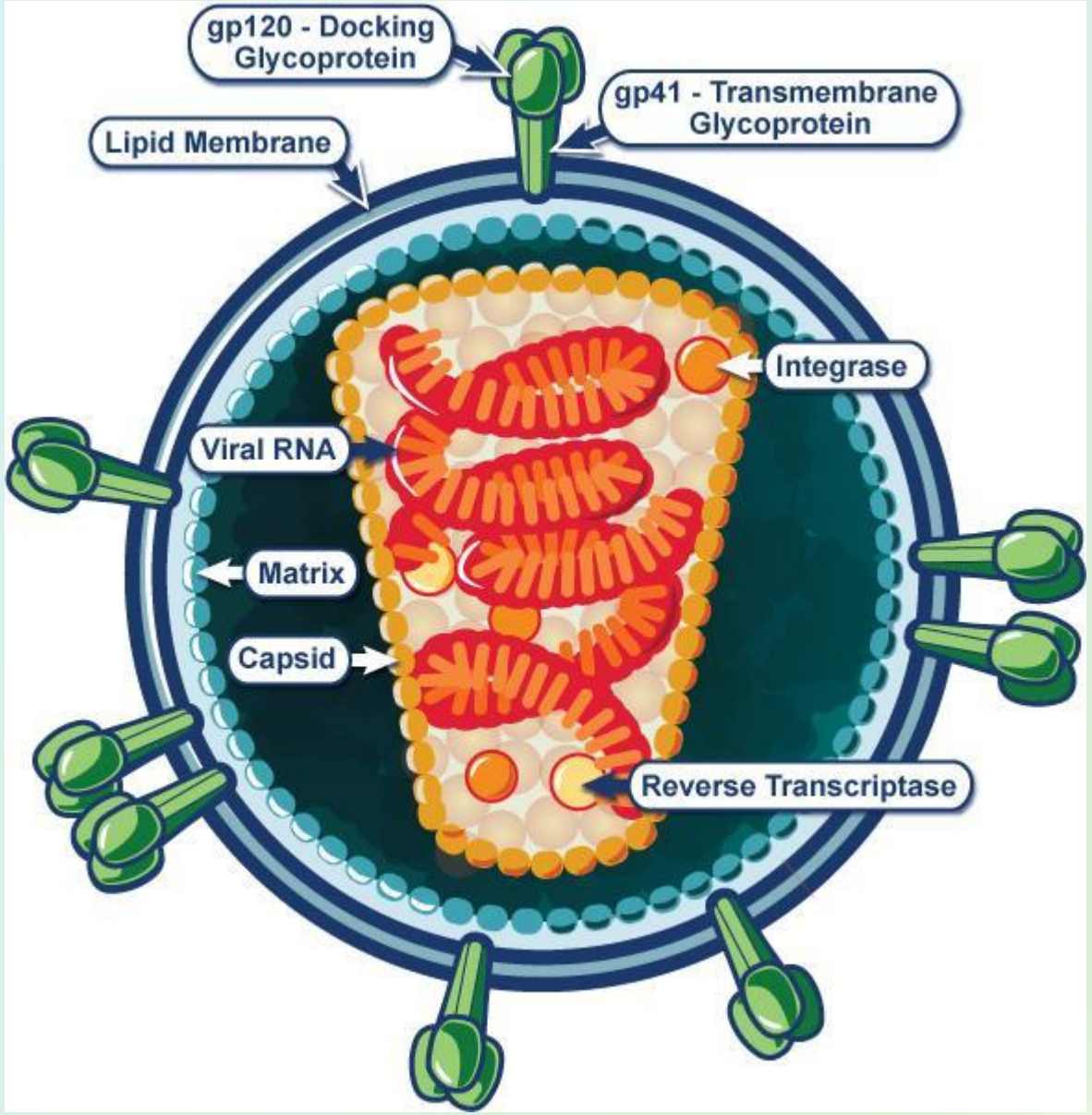
1. Through exposure to infected blood:
    1. Sharing needles (common in drug addicts)
    2. Blood transfusion with infected blood
2. Healthcare workers:
    1. Risk via needle-stick injuries

Perinatal (Vertical) Transmission

1. From mother to child
2. Occurs: During pregnancy, At delivery, Through breastfeeding
3. Risk without treatment: Approximately 40%
4. Key factor: Maternal viral load at delivery

Morphologic Structure of HIV-1

1. Human Immunodeficiency Virus Type 1 belongs to the lentivirus subgroup of retroviruses.
2. Diameter: ~100 nm
3. Shape: Spherical, enveloped virus
4. Surrounded by a lipoprotein envelope derived from host cell membrane



1. Envelope (Outer Layer): Lipid bilayer obtained from host cell during budding Contains 72 glycoprotein spikes

Important proteins:

1. **gp120 (surface glycoprotein)** : Responsible for attachment (adhesion) to CD4 receptors on host cells. Loosely attached → easily shed into serum & lymphatic tissues
2. **gp41 (transmembrane glycoprotein)** : Anchors gp120 to viral envelope. Helps in fusion of virus with host cell membrane
3. May incorporate host proteins: **HLA class I & II**
4. Adhesion molecules like **ICAM-1**

2. Matrix Layer: Located just beneath the envelope, Composed of: p17 protein

Functions:

1. Maintains viral structure
2. Anchors envelope to capsid

3. Capsid (Core) Cone-shaped central structure

Made of: **p24 protein (core antigen):** p24 is an important diagnostic marker in early HIV infection

4. Nucleocapsid: Inside the capsid

1. Two copies of single-stranded RNA
2. RNA is associated with: p7 nucleoprotein

5. Viral Genome: Contains 2 identical strands of RNA . Carries genetic information required for replication

6. Enzymes (Essential for Replication) HIV contains enzymes necessary for its life cycle:

1. **Reverse Transcriptase (p66)** : Converts viral RNA → DNA
2. **Integrase (p32)** : Integrates viral DNA into host genome
3. **Protease (p11)** Cleaves viral proteins → maturation of virus

7. Additional Features

1. During budding, virus may carry **host cell components**
2. gp120 shedding contributes to:
    1. Immune evasion
    2. Circulating viral proteins in blood

1. **gp120** → Attachment to CD4 cells
2. **gp41** → Fusion with host membrane
3. **p17** → Structural matrix protein
4. **p24** → Core antigen (diagnostic marker)
5. **p7** → RNA binding protein
6. **RT, Integrase, Protease** → Replication machinery

Pathophysiology

HIV Exposure

↓

Binding to CD4 T-cell (gp120 + CD4 receptor)

↓

Co-receptor Binding (CCR5 / CXCR4)

↓

Fusion with Host Cell (gp41 mediated)

↓

Release of Viral RNA + Enzymes into Cytoplasm

↓

Reverse Transcription
(RNA → Double-stranded DNA by Reverse Transcriptase)

↓

Integration into Host Genome
(by Integrase enzyme)

↓

Transcription & Translation
(using host cell machinery)

↓

Formation of Immature Viral Particles

↓

Budding from Host Cell

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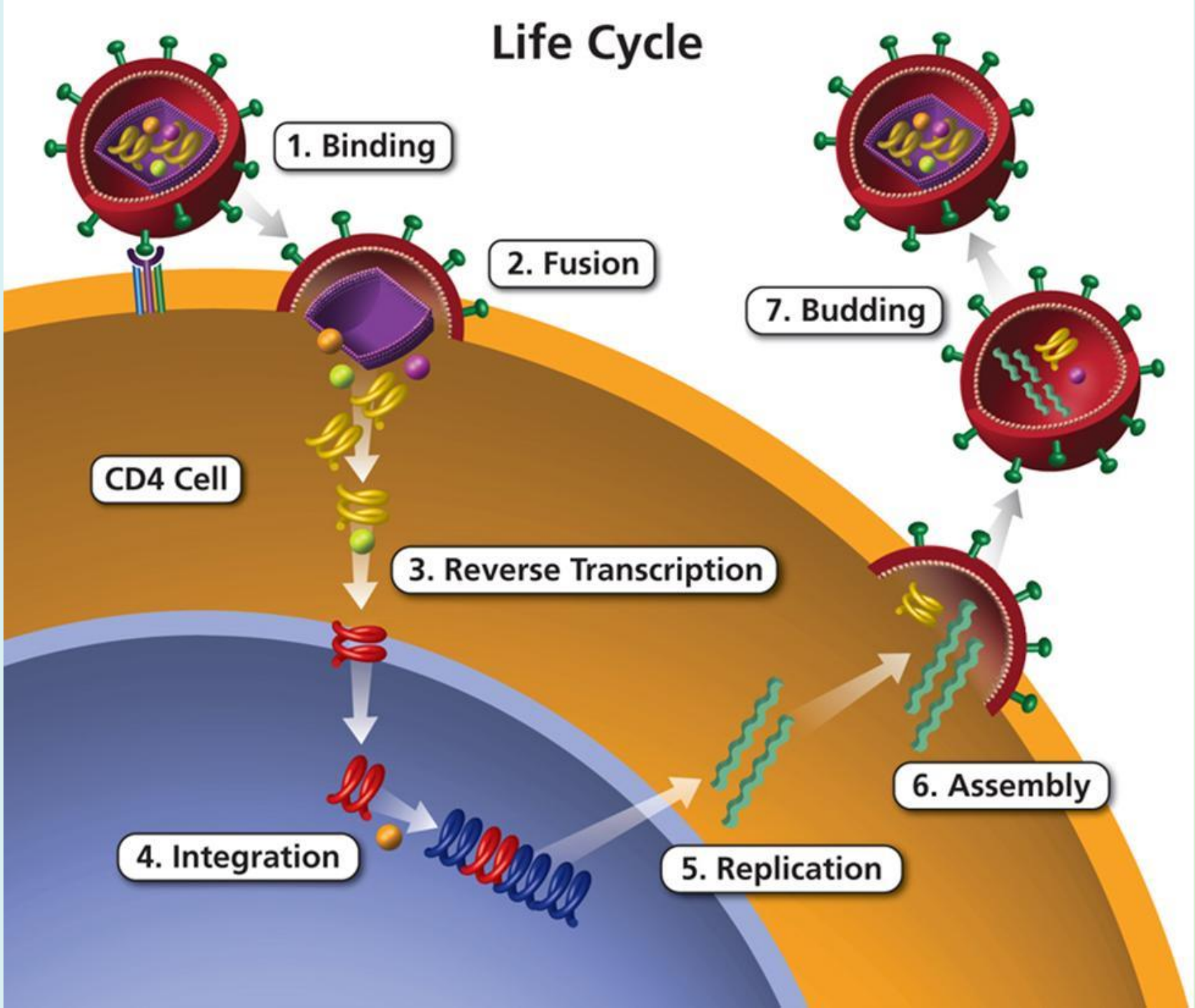
Maturation of Virus (Protease action)

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New Infectious Virions Released

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Infection of New CD4 Cells


Life Cycle
1. Binding
2. Fusion
CD4 Cell
3. Reverse Transcription
4. Integration
5. Replication
6. Assembly
7. Budding

1. Cell may be destroyed due to: Direct viral damage (**cytolysis**) , Immune attack (**cytotoxic T-cells**) Natural cell death
2. **Viral Distribution ~99% of HIV** is present in: **Lymph nodes,** Immune-rich tissues
3. **Acute Phase (Acute Retroviral Syndrome) :** Often misdiagnosed or unnoticed

Initial Infection
↓
Rapid Viral Replication (High Viremia)
↓
Drop in CD4 Cells
↓
Immune Response Begins (Antibody Production)
↓
Symptoms Appear:
Fever
Rash
Lymphadenopathy
Fatigue
Night sweats

Viral Set Point Phase (Chronic Phase)

- Immune System Partially Controls Virus

↓

Balance Achieved:
Virus Produced = Virus Destroyed

↓

Establishment of "Viral Set Point"

- Higher set point → Faster disease progression
- Lower set point → Slower progression

Clinical Presentation

1. Primary (Acute) HIV Infection

Common early symptoms (Acute Retroviral Syndrome):

a. **Fever**
b. **Pharyngitis (sore throat)**
c. **Lymphadenopathy (adenopathy)**

Other associated features:

a. Rash
b. Fatigue
c. Night sweats

Disease Progression : Progression to AIDS depends on: **Viral RNA load**

- Higher viral load → **Faster progression to AIDS**

Clinical Features in Children: Most HIV-infected children are **asymptomatic initially**

Common physical findings:

1. Lymphadenopathy
2. Hepatomegaly
3. Splenomegaly
4. Failure to thrive
5. Weight loss / low birth weight
6. Fever of unknown origin

Diagnosis

1. **ELISA Test (Screening)** Detects: Antibodies against HIV-1 , Highly sensitive and specific

   False Positive may occur in: Multiparous women , Recent infections: Hepatitis B , Influenza and After **rabies vaccination**

   False Negative: Early infection (window period)

2. **Western Blot Assay**

3. **Viral Load Testing:** Measures: Amount of HIV RNA (viremia)

   Methods:

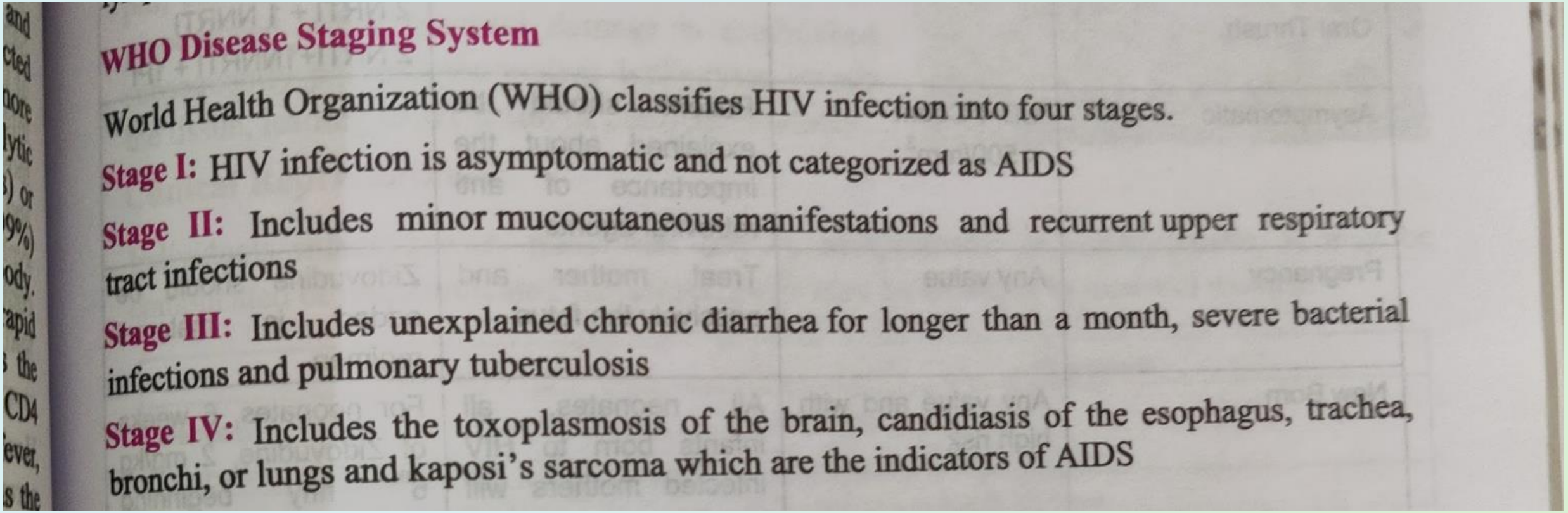
   1. RT-PCR (Reverse Transcriptase PCR)
   2. Branched DNA assay
   3. Nucleic acid sequence-based amplification

4. **CD4 Count** Normal range: 500 – 1600 cells/µL

   Important marker for:

   1. Disease progression
   2. Immune status

WHO Disease Staging System



WHO Disease Staging System

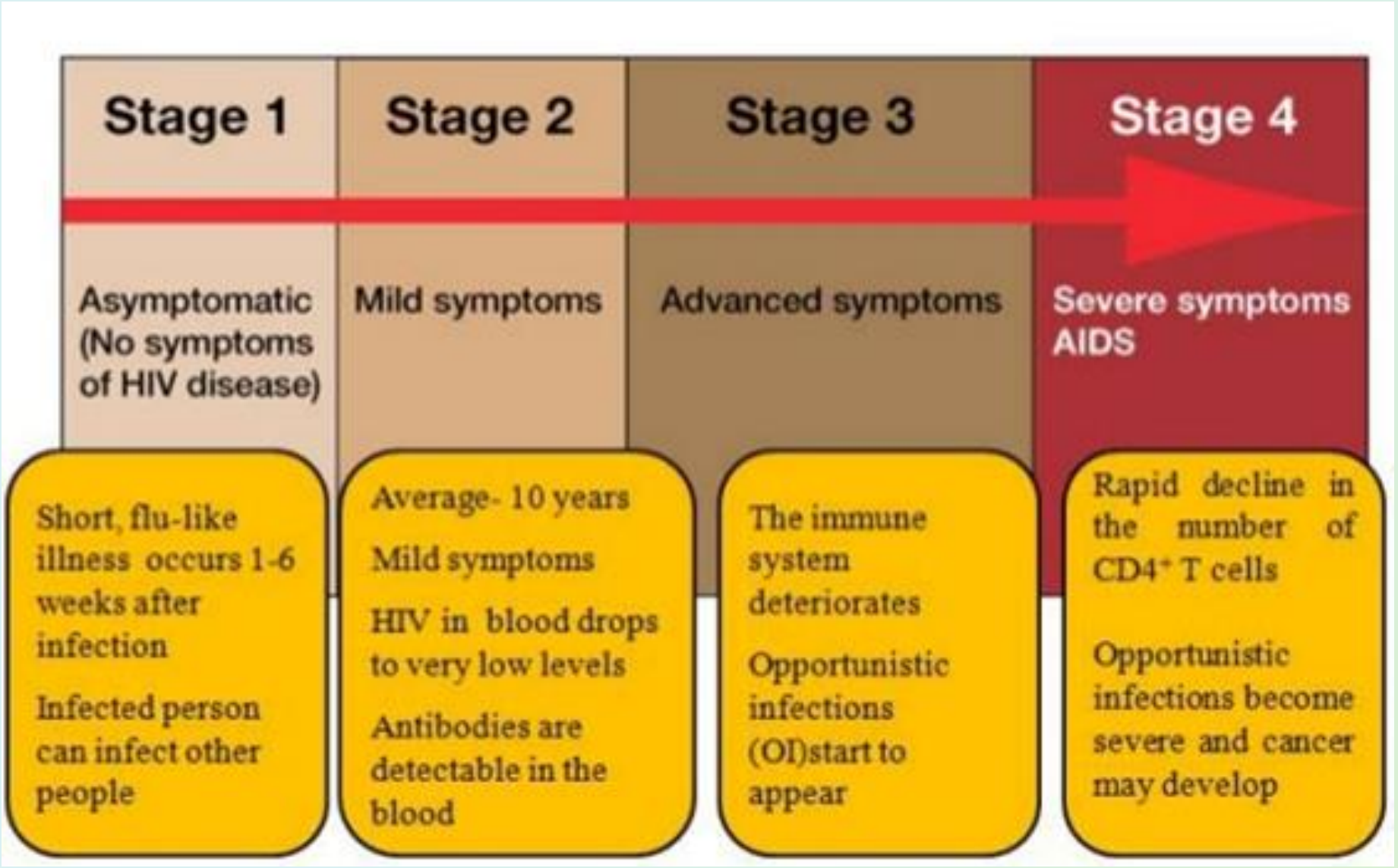
World Health Organization (WHO) classifies HIV infection into four stages.

Stage I: HIV infection is asymptomatic and not categorized as AIDS

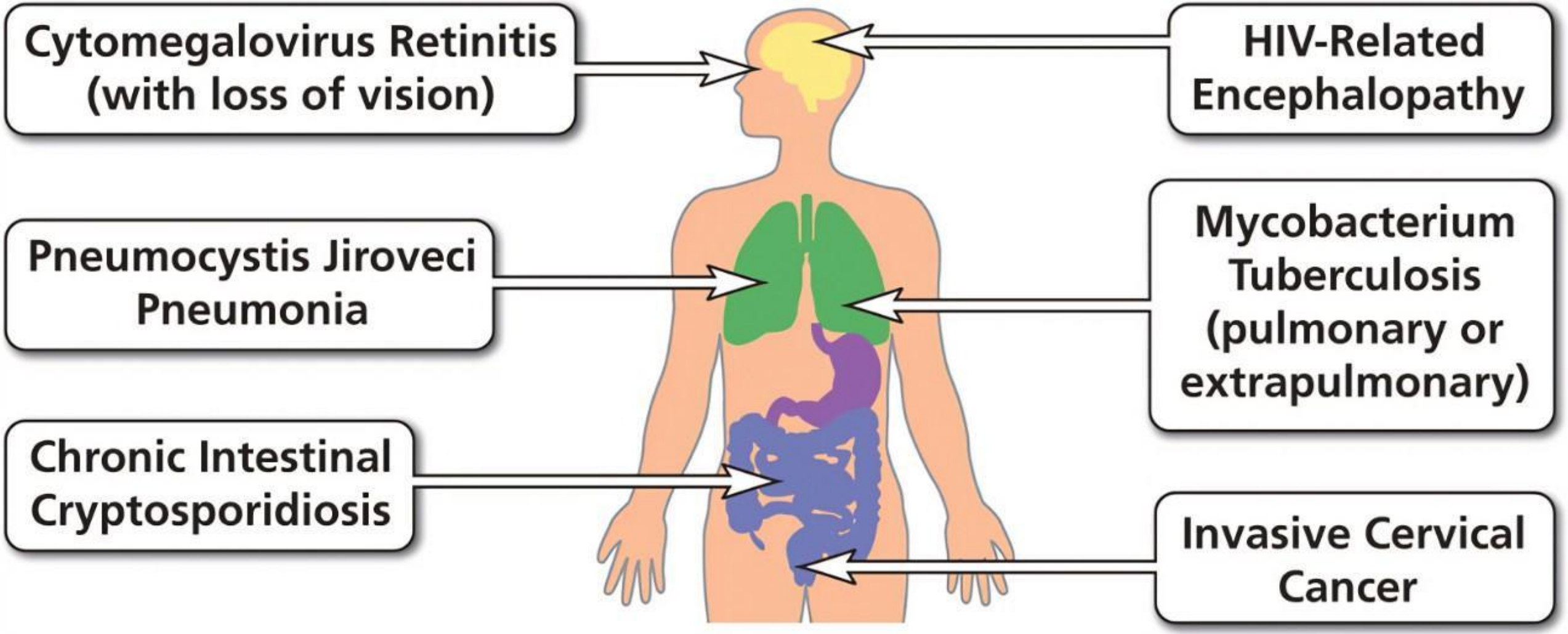
Stage II: Includes minor mucocutaneous manifestations and recurrent upper respiratory tract infections

Stage III: Includes unexplained chronic diarrhea for longer than a month, severe bacterial infections and pulmonary tuberculosis

Stage IV: Includes the toxoplasmosis of the brain, candidiasis of the esophagus, trachea, bronchi, or lungs and kaposi's sarcoma which are the indicators of AIDS


Stage 1
Stage 2
Stage 3
Stage 4
Asymptomatic (No symptoms of HIV disease)
Mild symptoms
Advanced symptoms
Severe symptoms AIDS
Short, flu-like illness occurs 1-6 weeks after infection
Infected person can infect other people
Average- 10 years
Mild symptoms
HIV in blood drops to very low levels
Antibodies are detectable in the blood
The immune system deteriorates
Opportunistic infections (OI)start to appear
Rapid decline in the number of CD4+ T cells
Opportunistic infections become severe and cancer may develop

Examples of **AIDS-Defining Conditions**



Treatment
Non-Pharmacological

1. General Principle HIV-infected individuals may remain **asymptomatic** when:
CD4+ count is within normal range (500–1600 cells/µL)

Goal: Maintain **immune strength** , Delay disease progression

2. Lifestyle Modifications

a. Practice safe sexual behavior: Limit to one sexual partner , Consistent condom use
b. Avoid exposure to infection: Do not share blades, razors, needles, or sharps
c. Maintain personal hygiene: Regular handwashing, Clean surroundings

3. Dietary Measures

Consume a nutritious, balanced diet, Well-cooked foods . Properly washed fruits and vegetables

Avoid: Contaminated or raw foods (to prevent infections)

4. Avoid Harmful Habits: Abstain from alcohol, Avoid smoking/tobacco , Avoid substance abuse

5. Infection Prevention Stay away from: Crowded or unhygienic environments and Situations increasing infection risk, Early management of minor infections

6. Overall Health Maintenance: Adequate rest and sleep, Regular physical activity (moderate), Stress management

Treatment
Pharmacological

- **1. Goal of Therapy Prevent viral replication**
  - Ultimate target:
    - **Undetectable viral load**
  - Benefits:
    - Improves **CD4+ count**
    - Reduces disease progression
    - Prevents opportunistic infections

- **2. Guidelines**
  - Recommended by:
    - International AIDS Society
    - Centers for Disease Control and Prevention
    - World Health Organization

Class	Full Form	Mechanism of Action	Examples
NRTIs	Nucleoside Reverse Transcriptase Inhibitors	Inhibit reverse transcriptase enzyme and block conversion of **RNA → DNA**	Zidovudine, Didanosine, Stavudine, Zalcitabine, Lamivudine, Abacavir
NNRTIs	Non-Nucleoside Reverse Transcriptase Inhibitors	Directly inhibit reverse transcriptase enzyme (**non-competitive inhibition**)	Nevirapine, Delavirdine, Efavirenz
PIs	Protease Inhibitors	Inhibit protease enzyme and prevent formation of **mature viral particles**	Saquinavir, Ritonavir, Indinavir, Nelfinavir, Amprenavir

Clinical category	CD4 T cell count and HIV viral load	Treatment recommendations	Treatment regimens
Symptomatic Unexplained fever Oral Thrush	Any value	Treat	2 NRTI + 1 PI 2 NRTI+ 2 PI 2 NRTI + 1 NNRTI 2 NRTI+1NNRTI + 1PI
Asymptomatic	CD 4 Count < 500/mm^3	Patient should be explained about the importance of anti retroviral agents	
Pregnancy	Any value	Treat mother and passively the fetus	Zidovudine should be added in maternal regimen
New Born	Any value and with high risk	All neonates, all infants born to HIV infected mothers will be HIV +	For neonates, 6 weeks of Zidovudine, 2 mg/kg 6th hrly beginning 1-8 hours of birth
Occupational and non occupational exposure prophylaxis		Treatment should be initiated within 72 hr of exposure	2 NRTI+ PI

Patient Education

About the Disease

1. HIV is a **chronic manageable infection**, not a death sentence
2. Caused by Human Immunodeficiency Virus which weakens the immune system
3. Progresses slowly if untreated → may lead to AIDS
4. Transmission occurs via:
    1. Unprotected sexual contact
    2. Blood and needle sharing
    3. Mother-to-child transmission
5. Patients may remain **asymptomatic for years**

About the Medication (ART)

a) Antiretroviral therapy (ART):
   a) **Controls viral replication**
   b) Helps maintain/increase **CD4 count**

b) Key counseling points:
   a) **Strict adherence is essential** (take medicines daily, same time)
   b) Missing doses → **drug resistance & treatment failure**
   c) Do not stop medication without consulting doctor
   d) Be aware of possible **side effects** and report them

c) Goal:
   a) Achieve **undetectable viral load**

3. Lifestyle Modifications

A. Prevention of Transmission

- Use condoms consistently
- Maintain single sexual partner
- Do not share:
  - Needles
  - Razors/blades

B. Healthy Living

- Eat nutritious, well-cooked food
- Maintain personal hygiene
- Regular exercise and adequate rest

C. Avoid Harmful Habits

- No alcohol
- No smoking/tobacco
- Avoid substance abuse

D. Infection Prevention

- Avoid:
  - Unhygienic environments
  - Contaminated food/water
- Seek early treatment for infections

E. Psychosocial Support

- Maintain positive mental health
- Seek family and social support
- Avoid stigma and discrimination

Feature	HIV Infection	AIDS (Acquired Immunodeficiency Syndrome)
Definition	Infection caused by Human Immunodeficiency Virus	Advanced stage of HIV infection
Stage	Early to chronic phase	Final/advanced stage
Immune Status	CD4 count may be normal or mildly reduced	**Severely reduced CD4 (<200 cells/mm³)**
Symptoms	Often asymptomatic or mild (fever, rash, lymphadenopathy)	Severe symptoms due to opportunistic infections
Opportunistic Infections	Usually absent	Common (TB, candidiasis, toxoplasmosis, etc.)
Viral Load	Variable, can be controlled with ART	High if untreated
Diagnosis	HIV antibody/antigen tests (ELISA, etc.)	HIV positive + AIDS-defining conditions
Reversibility	Can be controlled with ART	Cannot be reversed, but can be managed
Prognosis	Good with treatment	Poor if untreated